

Homework #11

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Question 5.

a.

Base case: $n = 1$.

For $n = 1$,

$$n^3 + 2n = 1^3 + 2 * 1 = 3$$

Thus, 3 divides $n^3 + 2n$ for $n = 1$.

Induction: For all $k \geq 1$, if 3 divides $k^2 + 2k$, we prove 3 divides $(k + 1)^2 + 2(k + 1)$.

$$k^3 + 2k = 3m, m \in \mathbb{Z}^+$$

$$(k + 1)^3 + 2(k + 1) = k^3 + 3k^2 + 3k + 1 + 2k + 2$$

$$= (k^3 + 2k) + (3k^2 + 3k + 3)$$

$$= 3m + 3(k^2 + k + 1)$$

$$= 3(m + k^2 + 2k + 1)$$

Since $m + k^2 + 2k + 1$ is an positive integer, 3 divides $(k + 1)^2 + 2(k + 1)$.

b.

Base case: $n = 2$

For $n = 2$, 2 is a prime.

Induction: Assume for all $k \geq 2$, all integers can be written as a product of prime.

We then prove $k + 1$ can be written as a product of prime.

If $k + 1$ is prime, then it can be written as $1 * (k + 1)$

If $k + 1$ is not prime, then it can be written as

$$k + 1 = a * b \text{ for } 2 \leq a, b \leq k$$

Since we assumed for all $k \geq 2$, all integers can be written as a product of prime.

Thus, a and b are products of prime.

Thus, $k + 1$ is a product of prime.

Question 6.

(a): 7.4.1

A.

$P(3)$ is true since $1^2 + 2^2 + 3^2 = (3(3+1)((3*2)+1))/6 = 14$.

B.

$$P(k) = \sum_{j=1}^k j^2 = \frac{k(k+1)(2k+1)}{6}$$

C.

$$P(k+1) = \sum_{j=1}^{k+1} j^2 = \frac{(k+1)(k+1+1)(2(k+1)+1)}{6}$$

D.

$P(1) = \frac{6}{6} = 1$. Thus $P(1)$ is true.

E.

We prove in the inductive step that: $\sum_{j=1}^{k+1} j^2 = \frac{(k+1)(k+1+1)(2(k+1)+1)}{6}$
For all $k \geq 1$, $P(k)$ implies $P(k+1)$.

F.

The inductive hypothesis is $P(k) = \sum_{j=1}^k j^2 = \frac{(k+1)(2k+1)}{6}$

G.

Base case: $n = 1$

$$\sum_{j=1}^1 k^2 = 1^2 = 1$$

$$1 * (1+1) * (2 * 1 + 1) / 6 = (1 * 2 * 3) / 6 = 1$$

Thus, $P(1)$ is true.

Induction: For all $k \geq 1$, if $P(k)$ is true, we prove $P(k+1)$ is also true.

$$\sum_{j=1}^{k+1} (k+1)^2 = \sum_{j=1}^k k^2 + (k+1)^2$$

$$= \frac{k(k+1)(2k+1)}{6} + (k+1)^2$$

$$= \frac{k(k+1)(2k+1)}{6} + \frac{6(k+1)(k+1)}{6}$$

$$= \frac{(k+1)(2k^2+7k+6)}{6}$$

$$= \frac{(k+1)(k+2)(2k+3)}{6}$$

$$= \frac{(k+1)((k+1)+1)(2(k+1)+1)}{6}$$

$$= \frac{l(l+1)(2l+1)}{6} \text{ for } l = k+1$$

Thus, for all $k \geq 1$, $P(k+1)$ is true if $P(k)$ is true.

(b):7.4.3

Prove that for $n \geq 1$: $\sum_{j=1}^n \frac{1}{j^2} \leq 2 - \frac{1}{n}$.

Base case: $n = 1$:

$$f(1) = \frac{1}{1^2} \leq 2 - \frac{1}{1} = 1 \leq 1$$

Base case is true.

Induction:

We assume

$$\sum_{j=1}^k \frac{1}{j^2} \leq 2 - \frac{1}{k}$$

We prove that:

$$\sum_{j=1}^{k+1} \frac{1}{j^2} \leq 2 - \frac{1}{k+1}$$

$$\begin{aligned} \sum_{j=1}^{k+1} \frac{1}{(k+1)^2} &= \sum_{j=1}^k \frac{1}{k^2} + \frac{1}{(k+1)^2} \\ &\leq 2 - \frac{1}{k} + \frac{1}{(k+1)^2} \\ &\leq 2 - \frac{1}{k} + \frac{1}{k(k+1)} \\ &\leq 2 - \frac{(k+1)}{k(k+1)} + \frac{1}{k(k+1)} \\ &\leq 2 - \frac{k}{k(k+1)} \\ &\leq 2 - \frac{1}{(k+1)} \end{aligned}$$

Thus, for all $k \geq 1$, $P(k+1)$ is true if $P(k)$ is true.

(c):7.5.1

Base case: $n = 1$

For $n = 1$, $3^{2*1} - 1 = 8 = 2 * 4$

Thus, $P(1)$ is true.

Induction: For all $k \geq 1$, we prove $P(k+1)$ is true if $P(k)$ is true.

$$3^{2k} - 1 = 4m, m \in \mathbb{Z}^+$$

$$3^{2(k+1)} - 1 = 3^{2k+2} - 1$$

$$= 3^2 * 3^{2k} - 1$$

$$= 9 * 3^{2k} - 9 + 8$$

$$= 9 * (3^{2k} - 1) + 8$$

$$= 9 * 4m + 8$$

$$= 4 * (9m + 2)$$

Since $9m + 2$ is a positive integer, 4 evenly divides $3^{2(k+1)} - 1$. Thus, for all $k \geq 1$, $P(k+1)$ is true if $P(k)$ is true.